



Boron cross-linked graphene oxide/polyvinyl alcohol nanocomposite gel electrolyte for flexible solid-state electric double layer capacitor with high performance



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ARTICLE INFO

Article history:

Received 8 January 2014
Received in revised form 27 March 2014
Accepted 27 March 2014
Available online 12 April 2014

Keywords:

polymer nanocomposite gel electrolyte
graphene oxide
boron cross-linking
supercapacitor
electrochemical properties

ABSTRACT

A new family of boron cross-linked graphene oxide/polyvinyl alcohol (GO-B-PVA) nanocomposite gels is prepared by freeze-thaw/boron cross-linking method. Then the gel electrolytes saturated with KOH solution are assembled into electric double layer capacitors (EDLCs). Structure, thermal and mechanical properties of GO-B-PVA are explored. The electrochemical properties of EDLCs using GO-B-PVA/KOH are investigated, and compared with those using GO-PVA/KOH gel or KOH solution electrolyte. FTIR shows that boron cross-links are introduced into GO-PVA, while the boronic structure inserted into agglomerated GO sheets is demonstrated by DMA analysis. The synergy effect of the GO and the boron crosslinking benefits for ionic conductivity due to unblocking ion channels, and for improvement of thermal stability and mechanical properties of the electrolytes. Higher specific capacitance and better cycle stability of EDLCs are obtained by using the GO-B-PVA/KOH electrolyte, especially the one at higher GO content. The nanocomposite gel electrolytes with excellent electrochemical properties and solid-like character are candidates for the industrial application in high-performance flexible solid-state EDLCs.

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1. Introduction

Electric double layer capacitors (EDLCs), also referred to supercapacitors, are one type of very important energy storage devices with high specific capacity, high power density and long cycle-life.[1] An unmet challenge is to design and develop safer, more reliable and high-performance polymer gel electrolytes (PGEs) for miniaturization and ultra-thin design of flexible supercapacitors without leakage problems.[2–5] Polyvinyl alcohol (PVA) is a synthesized water-soluble polymer, and a candidate for water-based PGEs due to relatively low cost, environmental-friendliness and safety. Many PVA based PGE systems have been developed and shown promising results.[6–11] Crosslinking is essential to endow PGE with good mechanical flexibility. The in-situ crosslinking of PVA during the assembly of devices was chosen to reduce IR drop or to benefit for ion transportation, but it is a cumbersome

and time-consuming process.[12,13] Freeze-thaw method can lead to physically crosslink PVA hydrogels with a combination of suitable strength and toughness.[14,15] Such gel electrolytes can be sandwiched between electrodes for the fast assembly of EDLCs by using roll-to-roll technology, which is a reliable and large-scale production method.[16] However, maintaining the good contact between electrode and electrolyte is still a challenge.

On the other hand, it is worthy of enhancing the ionic conductivity of PVA hydrogel electrolyte to gain better performance of EDLC. Recently, the PGEs doped with electrically insulating graphene, e.g. graphene oxide (GO), have been applied in some electrochemical devices like dye-sensitized solar cells,[17] fuel cells[18,19] and supercapacitors[5] thanks to the high ionic conductivity ($2.1 \pm 0.2 \text{ S cm}^{-1}$) of GO. [20] GO can act as an ionic conducting promoter owing to surface polar groups and its conjugate structure facilitates transportation of cations. However, the steric effects of GO sheets (e.g. restacking, blocking) deteriorating ion conductivity when GO content is over 1wt% have been observed in recent reported literatures,[5,19] limiting GO to give full play to the role.

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Boron atoms can act as physical or chemical cross-links of the PVA matrix improving the thermal stability and the mechanical properties of GO-PVA nanocomposite. When boron-crosslinking GO-PVA nanocomposite, “boron” crosslinking structure can be formed through the dehydration between boric acid ($\text{B}(\text{OH})_3$) and hydroxyl groups ($-\text{OH}$) that exist in the PVA chains and GO sheets. [21–23] It is noted that the doping of boron within carbon electrodes for enhancing wettability of electrolyte-electrode interface and introducing pseudocapacitance has been studied previously.[24] However, the combination of boron crosslinking and GO incorporation with polymer electrolyte to improve performance of PGE and the electrode-electrolyte contact has not been explored yet.

In this study, a new family of the polymer nanocomposite gel electrolytes, which was based on GO incorporated into PVA hydrogels and boron cross-linked, was developed. Freeze-thaw cycling produced gel electrolytes with good mechanical properties for the convenient assembly of flexible EDLCs. The gel electrolytes was saturated with KOH solution before being assembled. The role of KOH, either in polymer electrolyte or liquid electrolyte, acts as electrolyte salt to provide moveable ions for the electrolyte systems. Both polymer and pure water are weak in ion conductivity and the main functionality of them is to help the dissociation of the electrolyte salt through solvation effect. Electrochemical properties of EDLCs using GO-PVA/KOH or GO-B-PVA/KOH electrolyte or KOH liquid electrolyte were investigated and compared. In addition, the thermal stability and mechanical properties of the nanocomposite were also studied. It is hoped that GO can act as a promoter for the ionic conductivity, while boron cross-links have a synergy effect on improving the interfacial interaction and unblocking the channels for the ion transportation by being inserted into agglomeration of GO sheets. The combination of improved electrical performance and their solid-like character would make these new nanocomposite electrolytes amenable to roll-to-roll processing for the flexible energy storage devices.

2. Experimental

2.1. Preparation of boron cross-linked graphene oxide/polyvinyl alcohol (GO-B-PVA) nanocomposite hydrogel

Graphite flakes (XF026, fixed carbon $\geq 99.90\%$, 100 mesh, XF Nano, Inc., China) were used to prepare graphite oxide by Hummer method.[25] A certain amount of graphite oxide was dispersed in 2.5 mL 0.5wt% aqueous ammonia solution, and then ultrasonically-dispersed for 20 min. Under magnetic stirring, 5 g PVA (alcoholysis degree $> 99\%$) aqueous solution (0.1 g/mL) was added dropwise and stirred for 24 h to obtain a homogenous GO-PVA solution. The prepared solution was poured into a 90 mm plastic Petri dish. The dish was frozen at -20°C for 10 h and thawed at room temperature for 2 h. The freeze-thaw cycles were repeated for 5 times to obtain the GO-PVA hydrogel. Then the hydrogel was peeled off the dish and immersed into a sufficient amount of boric acid/ammonia solution (1 mg/mL, $\text{pH} \sim 11$) for over 24 h to allow crosslinking to obtain cross-linked GO-B-PVA hydrogel. For comparison, hydrogels without GO (PVA and B-PVA) were prepared in the same way.

2.2. Preparation of activated carbon electrode

The mixture consisting of activated carbon (AC, $1800 \text{ cm}^2/\text{g}$), acetylene black (AB) and polytetrafluoroethylene (PTFE) with the mass ratio of 82: 10: 8 was milled to a solid-like black slurry with the help of an appropriate amount of ethanol. The solid slurry was pressed into a thin film having the thickness of about $100 \mu\text{m}$, cut in a regular shape ($1 \text{ cm} \times 1 \text{ cm}$), and pressed together with nickel

foam under 18 MPa to make an AC electrode. Then the electrode was placed in a vacuum oven at 100°C and dried for 12 h.

2.3. Assembly of electric double layer capacitor

The as-prepared GO-B-PVA hydrogel was immersed into 6 mol/L (6 M) KOH solution to get full swelling, then cut into $1 \text{ cm} \times 1 \text{ cm}$ in size. Before being used, the carbon electrodes were immersed into 6 M KOH solution, degassed in vacuum, to ensure good wetting. EDLCs (electrode/gel polymer electrolyte/electrode) were assembled by pressing all the elements together and tested. Two types of EDLCs with gel electrolytes, electrode/gel polymer electrolyte/electrode without separator (AC/Gel/AC, Type I) and electrode/separators/gel polymer electrolyte/separators/electrode (AC/S/Gel/S/AC, Type II), were assembled by pressing them together. $\sim 0.3 \text{ mm}$ filter paper was chosen as the separator if necessary. The structure of Type I supercapacitor is shown in Fig. 1. For comparison, EDLC using liquid electrolyte was prepared by totally immersing the electrodes with the separator into 6 M KOH solution.

2.4. Sample characterization

2.4.1. Degree of crosslinking

The degree of crosslinking was estimated by the extraction in a Soxhlet Extractor that used tetrahydrofuran (THF) to remove the non-covalently, cross-linked portion of the hydrogel.

2.4.2. Attenuation total reflection (ATR) infrared spectra

Infrared absorption spectra was obtained with a Fourier infrared spectrometer (FTIR, Tensor 27, Bruker) in ATR mode.

2.4.3. Dynamic mechanical analysis (DMA)

DMA was conducted with DMA25 (01 dB) Metravib instrument from 30°C to 100°C or 120°C at a heating rate of $5^\circ\text{C}/\text{min}$ and a frequency of 1 Hz.

2.4.4. Thermal stability

Thermal stability was evaluated by a thermogravimetric analyzer (TGA-Q50, TA Instrument) with a heating rate $10^\circ\text{C}/\text{min}$ from 100°C to 600°C .

2.4.5. Morphology

Scanning electron microscope (SEM, JSM-5400, JEOL) was used to observe the liquid nitrogen freeze-fractured sections of dried gels. GO sheets dispersed in the GO-B-PVA were observed by using transmission electron microscopy (TEM, 120 kV, FEI Tecnai G2 Spirit).

2.4.6. Mechanical properties

Tensile properties of hydrogels were measured by a microcomputer controlled electronic universal testing machine (CMT-6103, SANS Co. China) equipped with a 10 N force sensor, a gauge length 20 mm and a drawing rate $100 \mu\text{m}/\text{s}$. Samples were cut into $\sim 50 \text{ mm} \times 5 \text{ mm} \times 0.40 \text{ mm}$ specimens and mechanical testing was repeated for at least 5 times to ensure reproducibility. Compression measurements of hydrogels were carried out by a dynamic thermal mechanical analyzer (DMA, DMA25, 01 dB, Metravib instrument) equipped with compression clamps. Alkaline hydrogels were cut into specimens in size ($\sim 10 \text{ mm}$ in diameter and $3 \sim 4 \text{ mm}$ in length). Before being tested, each sample was preloaded at 0.1 N. Stress-strain measurements were taken at a compression rate of 0.5 N min^{-1} . Rheological properties of the hydrogels were measured at ambient temperature by an ARES-RFS controlled strain rheometer (TA Instruments), with a 25-mm-diameter para-plate attached to a transducer. The gap at the apex of the para-plate was set to be $2 \sim 3 \text{ mm}$.

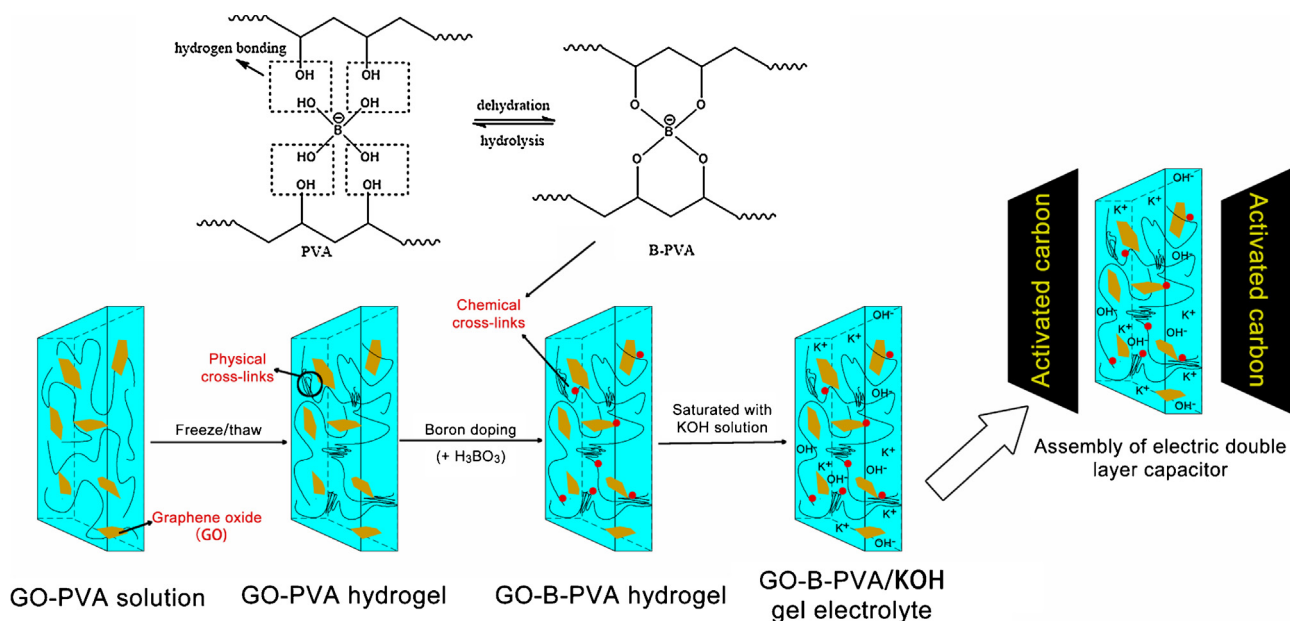


Fig. 1. Schematic representation of EDLC-Type I assembly along with the procedure of nanocomposite gel electrolyte synthesis.

2.5. Electrochemical measurements

The activated carbon electrodes were used as blocking electrodes. An electrochemical workstation (IM6e, Zahner Zennium, German) was used to measure cyclic voltammetry and electrochemical impedance spectroscopy (a frequency scan range: 10 mHz ~ 100 kHz, disturbed voltage: 5 mV). Charge-discharge curves and cycle life of supercapacitors were measured by using an Arbin supercapacitor test system.

3. Results and discussion

3.1. Structural characterization of GO-B-PVA nanocomposite gel electrolyte

Boronic structure incorporated into B-PVA and GO-B-PVA nanocomposite is investigated by FTIR, and the thermal stability of the composites is tested by TG. In Fig. 2a, characteristic absorption peak at 654 cm^{-1} corresponding to O-B-O bending and decreased intensity of O-H stretching vibration of ATR-FTIR spectra illustrates the boronic structure introduced into PVA chains.[26] The fact that much residue after extraction is obtained from another side proves the chemical crosslinking by boron doping. All the gels are controlled to have the same degree of crosslinking of ~97% according to the extraction results. Besides, as shown in Fig. 2b, the initial decomposing temperature of PVA is 264°C and increases to 306°C for the B-PVA and the GO-B-PVA nanocomposite owing to the boron cross-linked structure, while the GO incorporation has no effect on the thermal stability of B-PVA. To further investigate the interaction between the boron cross-links and/or the GO sheets within the PVA, glass transition temperature (T_g) related with activation energy for cooperative movement of polymer segments is studied by DMA. It can be seen from Fig. 3 that, B-PVA has a T_g much higher than that of PVA after the boron crosslinking, and T_g of 2wt% GO-B-PVA is 9°C higher than that of B-PVA. The improved T_g shows the movement of the polymer segments is inhibited by the boron cross-links and the GO layers, implying the insertion of the boron cross-links into the GO sheets for strengthening the interaction. Such structure of GO-B-PVA is hoped to be conducive to play the role of high ionic conductivity of GO.

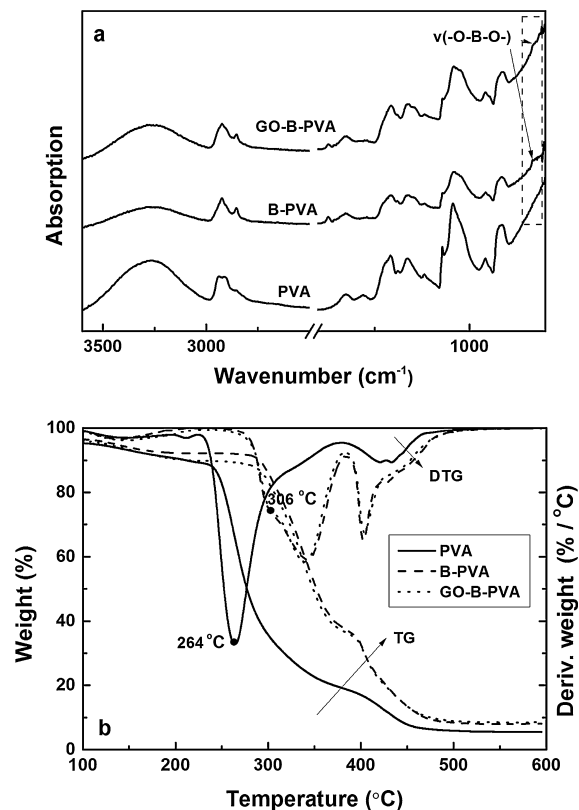


Fig. 2. (a) Attenuation total reflectance (ATR) infrared spectra of PVA, B-PVA and GO-B-PVA nanocomposite; (b) thermal stability of PVA, B-PVA and GO-B-PVA nanocomposite: the thermogravimetric (TG) curves and differential thermogravimetric (DTG) curves are marked.

The dispersion status of the GO sheets within the GO-B-PVA is observed by SEM and TEM. Fig. 4 shows the SEM images of (a) B-PVA, (b) 2wt% GO-B-PVA and (c) TEM image of 2wt% GO-B-PVA. From the SEM images we can see that morphology of fracture surface of GO-B-PVA nanocomposite looks rougher when

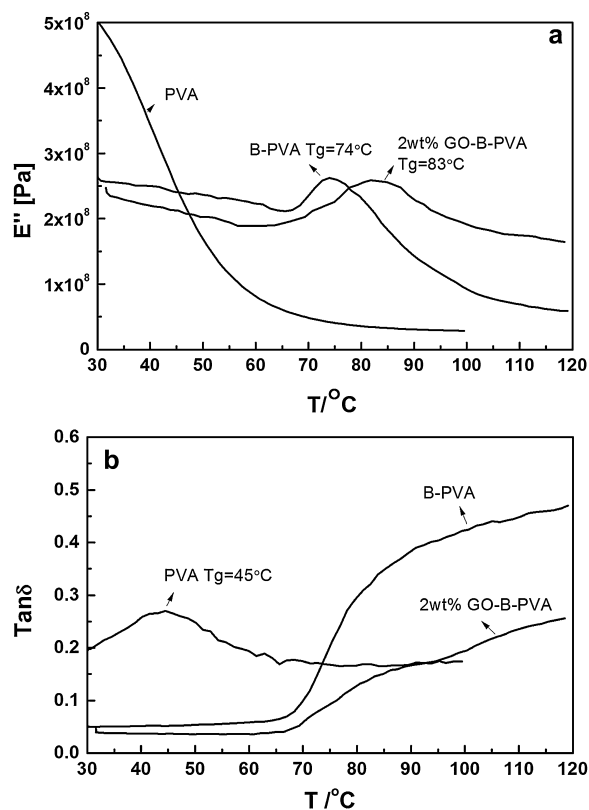


Fig. 3. DMA spectra of PVA, B-PVA and 2wt% GO-B-PVA nanocomposite: (a) loss modulus (E''); (b) $\tan \delta$.

incorporating GO sheets. In TEM image, the GO sheets are visible and have a good dispersion within the matrix.

3.2. Cyclic voltammetry (CV) of EDLCs with gel electrolyte

CV is often used to investigate electrochemical windows of electrodes or electrolyte. The electrodes can have redox properties to convert the energy between chemical energy and electric energy, but the electrolyte containing electrolyte salt (e.g. KOH) should be stable during working window for its role is just to provide mobile ions or act as an ion transport media. Therefore, the effects of boron cross-links, and/or GO sheets as well as PVA on the electrochemical stability of EDLC with KOH aqueous electrolyte are

investigated firstly. Fig. 5 shows CVs of EDLCs-type II with the KOH liquid electrolyte, with PVA/KOH, B-PVA/KOH and GO-B-PVA/KOH gel electrolytes under various voltage scan rates, respectively. The incorporation of PVA, and/or boronic structure, and/or GO doesn't make an adverse effect on the working windows of EDLCs with the KOH aqueous electrolyte. It confirms that all the electrochemical windows of the electrolytes are stable below 0.8 V. For the sweep rates from 10 mV/s to 200 mV/s, shapes of the CV curves of EDLCs are close to rectangular, indicating ideal capacitor characteristics. The shape of CV at higher sweep rates deviates from the rectangular due to Ohmic drop.[27]

By calculating the area of nearly rectangular CV curve, specific capacitance based on active carbon of a single electrode can be estimated as follows (1):

$$C = \frac{1}{sm\Delta V} \int_{V_0}^{V_0+\Delta V} i_d dV = \frac{1}{s\Delta V} \int_{V_0}^{V_0+\Delta V} i_d dV, \quad (1)$$

Where C is the specific capacitance (F/g), s is the voltage scan rate (V/s), m is the mass of active substance on the single electrode (g), V is the electric potential, i_d is the current density on the single electrode (A/g). In Fig. 6a, at the sweep rate of 10 mV/s, the area of CV related with specific capacity of GO-B-PVA/KOH gel electrolytes with various GO content and that of B-PVA in EDLCs-type II (with separator) are almost the same, which implies the ineffectiveness of GO incorporation when separating the direct contact between the electrolyte and the electrode; the results are much smaller than 130.4 F/g of KOH solution, e.g. the corresponding specific capacitance by using 2wt% GO-B-PVA/KOH gel electrolyte are only 103 F/g. However, the specific capacitance of EDLC-type I (without separator) with 2wt% GO-B-PVA/KOH gel electrolyte reaches 131 F/g, which is comparable to that of the device based on liquid electrolyte. In the inset of Fig. 6b, the specific capacitance by using 2wt% GO-B-PVA/KOH gel electrolyte increases to 226.2 F/g at the sweep rate of 1 mV/s. This value is much higher than 164 F/g of the EDLC with the liquid electrolyte. The superior of EDLC-type I to EDLC-type II devices based on gel electrolytes might be due to the excess capacitance originated from the contact interface between the electrode and the gel electrolyte or from lower Ohmic drop. To get more details about this, as shown in Fig. 6c, the Nyquist plots of EDLC-type I and EDLC-type II with 2wt% GO-B-PVA/KOH gel electrolyte having the same thickness to that of EDLC with KOH solution are investigated and compared. The general response of the plots is qualitatively described in terms of medium to high frequency semicircles associated with polymer electrolyte and charge transfer, followed by a low frequency line likely corresponding

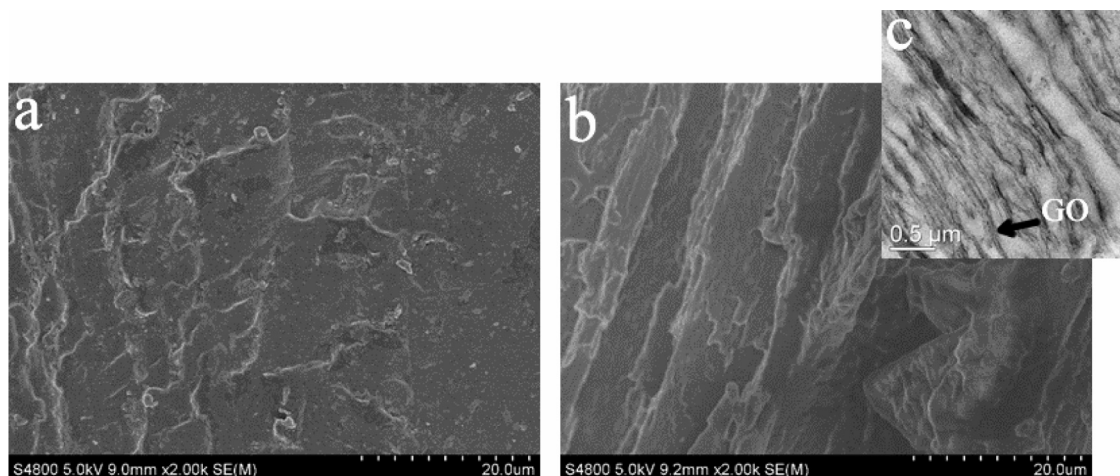


Fig. 4. Morphologies of (a) B-PVA and (b) 2wt% GO-B-PVA observed with scanning electron microscopy; (c) 2wt% GO-B-PVA observed with transmission electron microscopy.

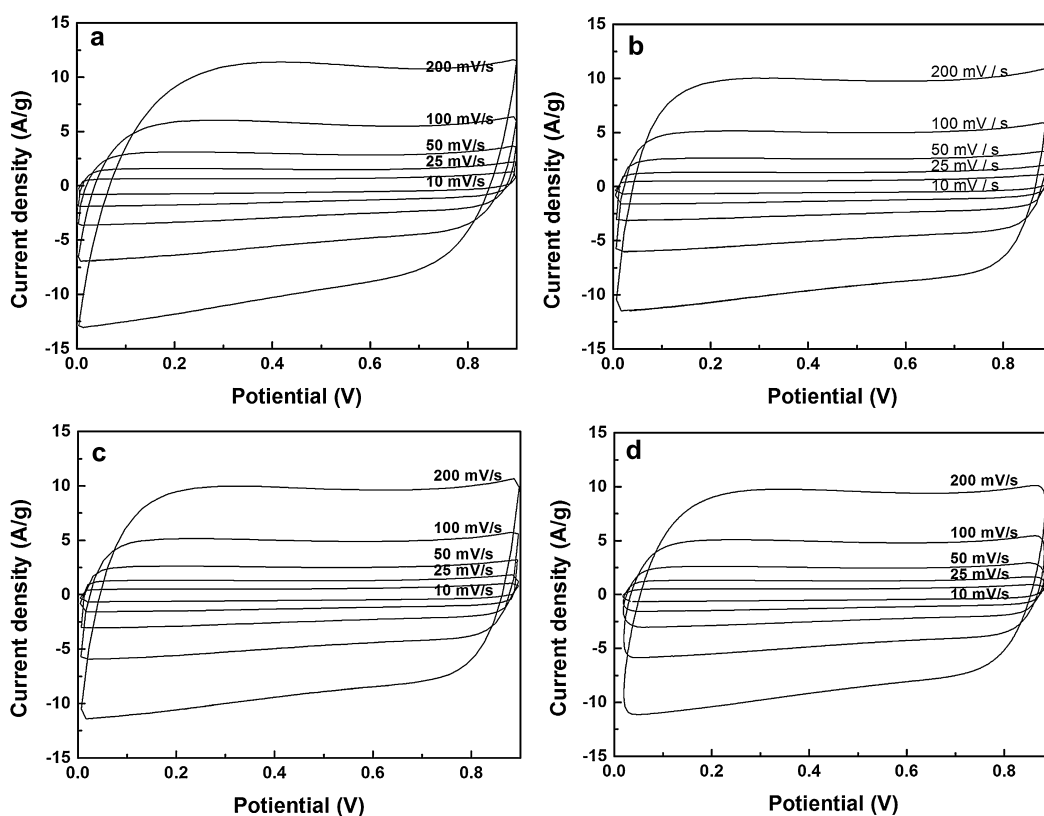


Fig. 5. Cycle voltammograms of EDLCs-type II assembled with (a) 6 M KOH aqueous solution; (b) PVA/KOH gel electrolyte; (c) B-PVA/KOH gel electrolyte and (d) GO-B-PVA/KOH gel electrolyte with filter paper as separators between electrode and electrolyte.

to EDLC capacitance. The bulk resistance (R_b) of electrolytes can be estimated from real part of impedance at high frequency of the semicircle, while the total resistance ($R_b + R_{ct}$) of bulk resistance and charge transfer resistance (R_{ct}) can be obtained from that at the medium frequency of the semicircle. For EDLC-type I, R_b and R_{ct} is 0.44 Ω and 0.34 Ω , respectively, lower than 0.55 Ω and 0.38 Ω of EDLC-type II. The dotted line at the low frequency range relates with the electrode-electrolyte contact. For the EDLC with the KOH solution, the slope of the impedance curve at the low frequencies is about 45°, close to that of Warburg impedance. On the other hand, for EDLCs with the gel polymer electrolytes, the slope is near 90°, suggesting the gel electrolyte may have a certain pseudo-capacitance. [13]

The analysis of the CV results demonstrates that GO sheets are potential in improving capacitance of EDLC by the suitable assembly, especially when the boron cross-links are introduced. Taking the advantage of devices without separator, Type I EDLCs with gel electrolyte are chosen for charge/discharge test under the working voltage window between 0 and 0.8 V to further investigate the effect of the boron cross-links and the GO sheets on the performance of EDLCs.

3.3. Charge/discharge behavior of EDLCs with gel electrolyte

Fig. 7 shows the discharge capacitance (specific capacitance based on active material on a single electrode) as a function of discharge current density (i_d) on a single carbon electrode, and the charge-discharge behaviors of the EDLCs at 0.6 A/g are shown in the insets. It is found that the discharge characteristics of EDLCs are almost linear, which confirms the capacitive behaviors of all studying EDLC cells. The discharge capacitance can be calculated

by galvanostatic charge-discharge cycle test according to the following formula (2):

$$C = 4 \times C_{cell} = 2 \times i_d \times t / \Delta V, \quad (2)$$

Where C_{cell} is the total capacitance of the EDLC cell, t is the discharge time (s), and ΔV is the potential difference (0.8 V). In Fig. 8a, EDLC with the KOH solution has a higher specific capacitance than that with GO-PVA/KOH gel electrolytes having no boron crosslinking at low current density, but its falling speed of the capacitance with increased i_d is larger. Increasing the content of GO seems to benefit for discharge performance. As i_d is larger than 1 A/g, 20wt% GO-PVA/KOH gel electrolyte performs competitively as the liquid electrolyte. The less deterioration of specific capacitance by using the GO-PVA/KOH gel electrolyte may be caused by the lower internal resistance of the electrode, which causes a smaller Ohmic drop at high discharging current density. In Fig. 7b, the specific capacitance of EDLCs with GO-B-PVA/KOH gel electrolytes outperforms that with liquid electrolyte when boron crosslinking is introduced into the gels. The discharge capacitance of EDLC by using 20wt% GO-B-PVA/KOH gel electrolyte at 0.1 A/g is 141.8 F/g, about a 129% increase compared to 109.4 F/g of that with the KOH aqueous solution electrolyte. An excess 29% increase of capacitance is probably originated from the GO sheets through the electrode-electrolyte contact. Fig. 7c shows the capacitance retention as a function of cycle number at 1 A/g. The capacitance is not only better for that with cross-linked gels but appears much more stable over repeated cycles (up to 1000 cycles).

3.4. Synergistic mechanism of GO and boron cross-links within the gel electrolyte

To better understand the function of the GO sheets and the boron cross-links on the electrochemical properties of gel

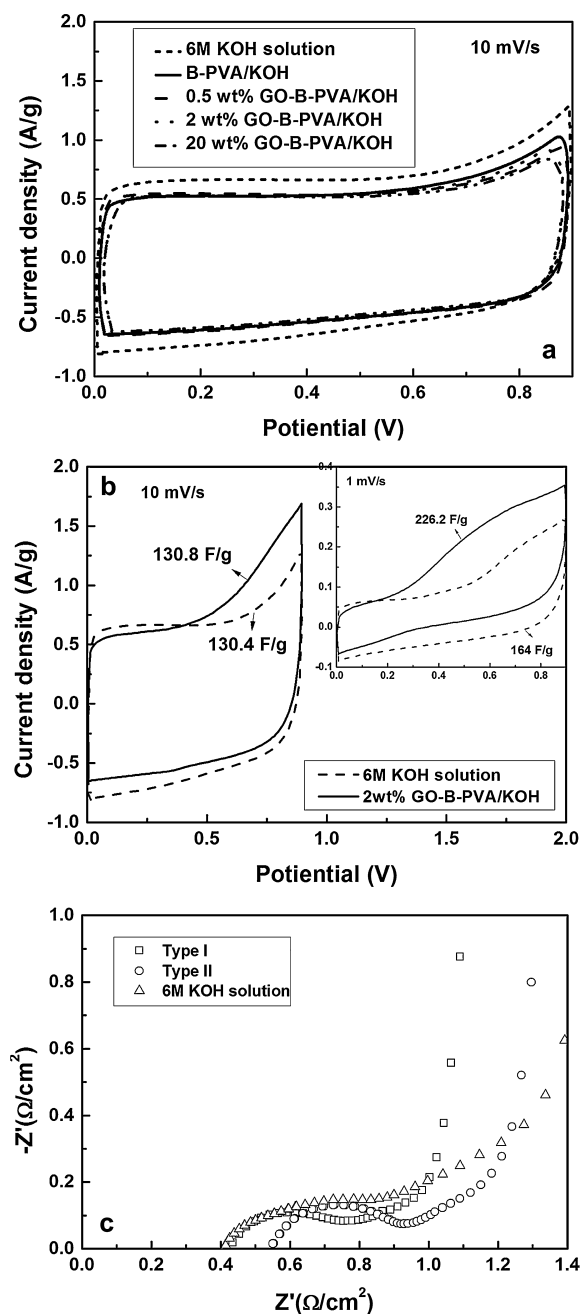


Fig. 6. (a) Influence of GO content on the cycle voltammograms (CVs) behavior of EDLCs-type II; (b) the CV of EDLC-type I with 2wt% GO-B-PVA/KOH gel electrolyte, compared with that of EDLC with 6 M KOH solution electrolyte; (c) electrochemical impedance spectra of 2wt% GO-B-PVA/KOH gel electrolyte superimposed with Type I and Type II and 6 M KOH solution.

electrolytes, the investigation on ionic conductivity of electrolytes is conducted. Higher ionic conductivity often contributes to smaller Ohmic drop during the working of EDLC.[28] The ionic conductivity (σ_c) of electrolytes can be calculated according to the following formula (3):

$$\sigma_c = \frac{L}{AR_b}, \quad (3)$$

where L is the thickness of the electrolyte, and A is the geometric area of the electrode/electrolyte contact (1 cm × 1 cm). The results of σ_c are shown in Fig. 8. The ionic conductivity of all of the gel electrolytes is better than that of the liquid electrolyte, and at the

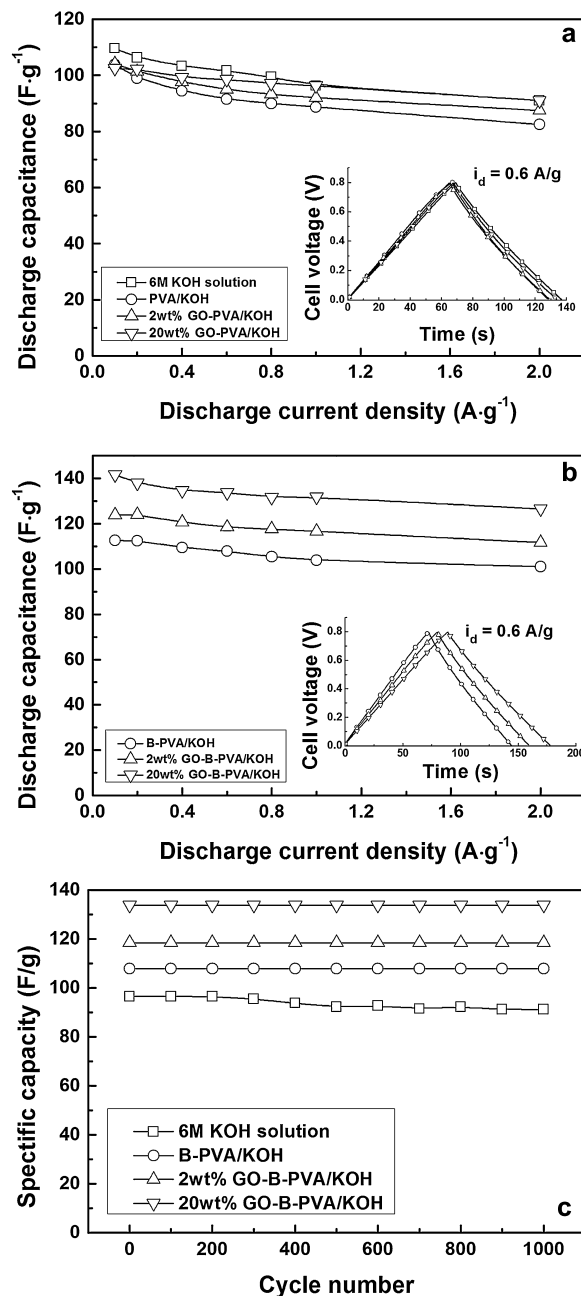


Fig. 7. Discharge capacitance as a function of discharge current density of EDLCs based on (a) GO-PVA/KOH gel electrolyte as well as KOH solution electrolyte and (b) GO-B-PVA/KOH gel electrolytes (the insets show charge-discharge curves of EDLCs at 0.6 A/g); (c) capacitance retention as a function of cycle number of EDLCs at 1 A/g.

low content of GO, the ionic conductivity also increases with the amount of GO. The improved conductivity is probably due to the presence of PVA chains and GO, both of which contain polar groups facilitating ionic transport. When the content of GO increases further, the ionic conductivity of GO-PVA/KOH gel decreases owing to the unavoidable aggregation of nanoparticles which leads to blocking ion channels or extending ion transport pathways, as illustrated in Fig. 9. The steric barrier effect on the ion transportation deteriorating the ion conductivity when GO content is over 1 wt% has been reported and still not overcome.[5,19] But in our studies when the boron cross-links is introduced into the GO-PVA nanocomposite, the ionic conductivity of 20wt% GO-B-PVA can reach 0.1950 S/cm, approximately 278% increase compared to 0.0702 S/cm of the liquid electrolyte. For one thing, boron atoms can accept anion or negative

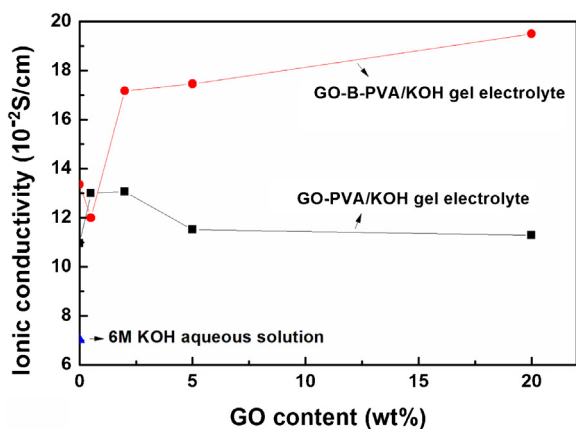


Fig. 8. Ionic conductivity of gel electrolytes as a function of GO content.

charge centers benefiting for the increase of the number of ions; for another thing, the cross-linked boronic structure could insert into the GO sheets and may open some of the blocked pathways for ion migration, which leads to the enhancement of tortuosity of ion transport pathways without extending the transporting distance. When boron-doping is executed on GO-PVA/KOH gel electrolytes, the unfavorable effect of GO aggregation on the ionic transportation is suppressed, and the synergy effect of GO and boron cross-links on enhancing ion conductivity is able to play. Therefore, the ionic conductivity of the GO-B-PVA/KOH gel electrolytes with the amount of GO over 2wt% can be further improved, gaining better performance for electrochemical devices.

3.5. Mechanical properties of gel electrolytes

For polymer gel electrolytes, apart from excellent ionic conductivity, the mechanical properties such as tension, compression and

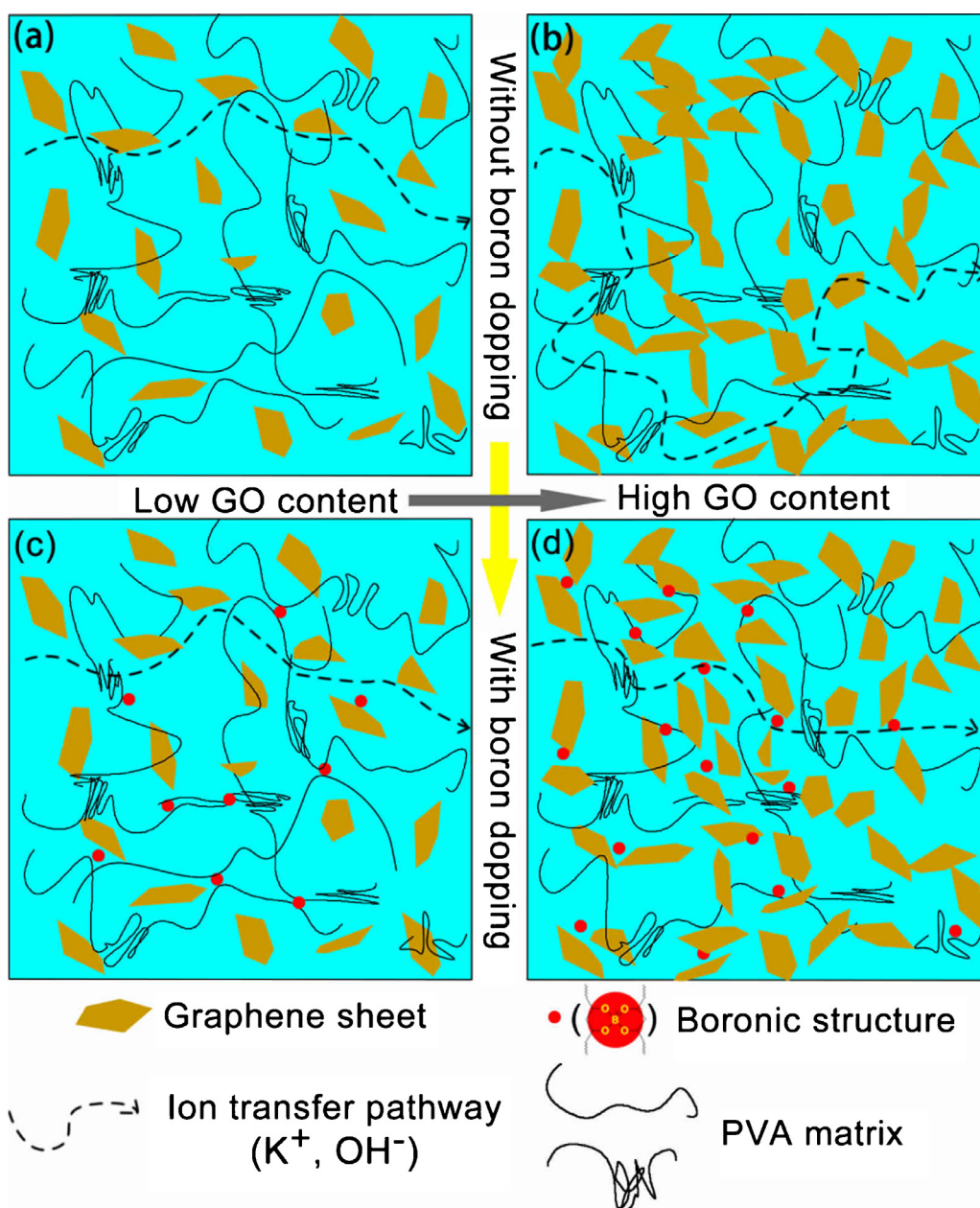


Fig. 9. Illustration of ion transport mechanism in GO-PVA and GO-B-PVA alkaline gel electrolytes with low and high GO content.

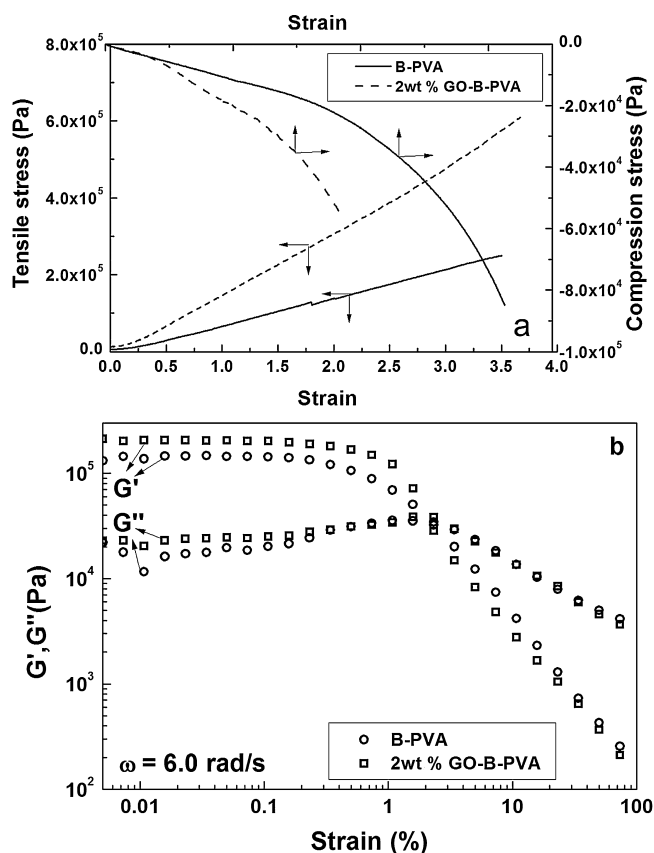


Fig. 10. Mechanical properties of gels: (a) tensile and compression properties; (b) shear property.

shear are necessary for the application in the assembly of flexible electrochemical devices. In Fig. 10, the cross-linked gels of B-PVA (without GO) exhibit a modulus of 0.073 MPa, tensile strength 0.25 MPa, and elongation to failure $\sim 350\%$. GO incorporation can improve the mechanical properties of the gels. The corresponding values of 2wt% GO-B-PVA gels are 0.15 MPa, 0.61 MPa, and 370%, respectively. Note that the modulus, tensile strength and elongation to failure all increase when GO is added. This simultaneous increase in mechanical properties is quite unusual for micron-composites.[29] Additionally, compression (compression strength -0.085 MPa, compression ratio $\sim 63\%$ for B-PVA; compression strength -0.056 MPa, compression ratio $\sim 37\%$ for 2wt% GO-B-PVA) and shear ($G' \sim 0.14$ MPa, $G'' \sim 0.022$ MPa for B-PVA; $G' \sim 0.21$ MPa, $G'' \sim 0.022$ MPa for 2wt% GO-B-PVA) tests are measured. Such hydrogels for gel electrolyte should be suitable for roll-to-roll (R2R) processing, which is an attractive technique for scale-up manufacturing of electrochemical devices.[7]

4. Conclusion

The boron cross-linked GO-B-PVA nanocomposite gel electrolyte was prepared by freeze-thaw/boron crosslinking method and then fully saturated with KOH solution. Structure of the GO-B-PVA nanocomposite was explored by FTIR and DMA, showing the incorporation of boronic structure into PVA matrix and insertion of such structure within the well-distributed GO sheets. Thermal stability was improved benefiting from the boron crosslinking. Electrochemical properties of EDLCs using gel and liquid electrolytes exhibiting the characteristics of ideal capacitors were investigated and compared. It was found that discharge

capacitance using GO-B-PVA/KOH gel electrolyte showed the synergistic effect of the GO sheets and the boron cross-links. With 20wt% GO contained, specific capacitance can be 141.8 F/g at 0.1 A/g, which increases up to 129.4% of that with the KOH aqueous solution electrolyte. Impedance spectroscopy studies showed that the gel electrolyte may have a certain pseudo-capacitance originated from interfacial contact. Higher ionic conductivity was gained for the gels at higher content of GO due to the increase of ion number and better ion transportation through the synergy effect of GO and the boron cross-links. The GO sheets can play a role of ionic promoters and cross-linked boronic structure could be inserted into the agglomerated GO sheets to open channels for ion migration. The good mechanical properties of GO-B-PVA endow the device with flexibility. It is believed that the GO-B-PVA gel electrolytes with good electrochemical properties, thermal stability and mechanical properties will be suitable for roll-to-roll technology, a scalable technique with a lot of potential for manufacturing of future energy storage devices, for assembly of flexible EDLCs with higher specific capacity and better cycle stability.

Acknowledgements

The authors are grateful for the support of the Natural Science Foundation of China (Grant: 51173207), Key projects of Guangdong Education Office (Grant: cxzd1101) and the Natural Science Foundation of Guangdong, China (Grants: 2011B090500004, 2012B091100313, 2012A090100006 and 2013C2FC0009). EPG acknowledges support from the Energy Materials Center at Cornell, an Energy Frontier Research Center funded by the U.S. Department of Energy, Office of Basic Energy Sciences, under Award No. DESC0001086. This publication is based on work supported in part by Award No. KUS-C1-018-02 from King Abdullah University of Science and Technology.

References

- [1] R. Kötter, M. Carlen, *Electrochim. Acta* 45 (2000) 2483.
- [2] A. Manuel Stephan, *Eur. Polym. J.* 42 (2006) 21.
- [3] J.Y. Song, Y.Y. Wang, C.C. Wan, *J. Power Sources* 77 (1999) 183.
- [4] D. Kalpana, N.G. Renganathan, S. Pitchumani, *J. Power Sources* 157 (2006) 621.
- [5] Y. Xi, F. Zhang, L. Zhang, T.F. Zhang, Y. Huang, Y.S. Chen, *Adv. Funct. Mater.* 23 (2013) 3353.
- [6] A.A. Mohamad, N.S. Mohamed, M.Z.A. Yahya, R. Othman, S. Ramesh, Y. Alias, A.K. Arof, *Solid State Ionics* 156 (2003) 171.
- [7] C.W. Huang, C.A. Wu, S.S. Hou, P.L. Kuo, C.T. Hsieh, H. Teng, *Adv. Funct. Mater.* 22 (2012) 4677.
- [8] J. Fu, J. Qiao, H. Lv, J. Ma, X.Z. Yuan, H. Wang, *ECS Trans* 25 (2010) 15.
- [9] V.M. Nikolic, A. Krkljes, Z.K. Popovic, Z.V. Lausevic, S.S. Miljanic, *Electrochem. Commun.* 9 (2007) 2661.
- [10] G. Merle, S.S. Hosseiny, M. Wessling, K. Nijmeijer, *J. Membr. Sci.* 409 (2012) 191.
- [11] Hu, X. L., Hou, G. M., Zhang, M. Q., Rong, M. Z., Ruan, W. H., Emmanuel P. Giannelis, *J. Mater. Chem.* 22 (2012) 18961.
- [12] A. Noda, M. Watanabe, *Electrochim. Acta* 45 (2000) 1265.
- [13] S. Nohara, H. Wada, N. Furukawa, H. Inoue, M. Morita, C. Iwakura, *Electrochim. Acta* 48 (2003) 749.
- [14] H.S. Choe, B.G. Carroll, D.M. Pasquariello, K.M. Abraham, *Chem. Mater.* 9 (1997) 369.
- [15] D.L. Zugic, I.M. Perovic, V.M. Nikolic, S.L. Maslovara, M.P.M. Kaninski, *Int. J. Electrochem. Sci.* 8 (2013) 949.
- [16] Z. Jiang, B. Carroll, K.M. Abraham, *Electrochim. Acta* 42 (1997) 2667.
- [17] Gun, Jenny, Sneha A. Kulkarni, Wang Xiu, Sudip K. Batabyal, Sergey Sladkevich, Petr V. Prikhodchenko, Vitaly Gutkin, Ovadia Lev, *Electrochem. Commun.* 19 (2012) 108.
- [18] Y.C. Cao, C. Xu, X. Wu, X. Wang, L. Xing, K. Scott, *J. Power Sources* 196 (2011) 8377.
- [19] Y.S. Ye, M.Y. Cheng, X.L. Xie, J. Rick, Y.J. Huang, F.C. Chang, B.J. Hwang, *J. Power Sources* 239 (2013) 424.
- [20] Stankovich, Sasha, Dmitriy A. Dikin, Richard D. Piner, Kevin A. Kohlhaas, Alfred Kleinhammes, Yuanyuan Jia, Yue Wu, SonBinh T. Nguyen, Rodney S. Ruoff, *Carbon* 45 (2007) 1558.

- [21] Tsukasa Miyazaki, Yuuki Takeda, Sachiko Akane, Takahiko Itou, Akie Hoshiko, Keiko En, *Polym* 51 (2010) 5539.
- [22] S. Hashimoto, K. Furukawa, *Biotechnol. Bioeng.* 30 (1987) 52.
- [23] Y.S. Zhu, X.J. Wang, Y.Y. Hou, X.W. Gao, L.L. Liu, Y.P. Wu, M. Shimizu, *Electrochim. Acta* 87 (2013) 113.
- [24] H. Guo, Q. Gao, *J. Power Sources* 186 (2009) 551.
- [25] W.S. Hummers Jr., R.E. Offeman, *J. Am. Chem. Soc* 80 (1958) 1339.
- [26] I. Uslu, H. Daştan, A. Altaş, A. Yayli, O. Atakol, M.L. Aksu, *e-Polymers* 7 (2013) 1568.
- [27] R. Chandrasekaran, Y. Soneda, J. Yamashita, M. Komada, H. Hotari, *J. Solid State Electrochem* 12 (2008) 386.
- [28] S.H. Wang, S.S. Hou, P.L. Kuo, H. Teng, *ACS Appl. Mater. Interfaces* 5 (2013) 8477.
- [29] S.Y. Fu, X.Q. Feng, B. Lauke, Y.W. Mai, *Composites Part B* 39 (2008) 933.